

Vivek Vishwanath

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EDUCATION

Georgia Institute of Technology

Graduation Date: May 2027

B.S./M.S. Computer Science

Major GPA: 4.0

- **Graduate Coursework:** Advanced Operating Systems, Advanced CPU Architecture, Programming Languages
- **Undergraduate Coursework:** Machine Learning, Compilers, Operating Systems, Data Structures & Algorithms
- **Concentrations:** Artificial Intelligence, Systems & Architecture

EXPERIENCE

State Farm - Life Insurance

May 2025 – August 2025

Software Engineer Intern

Dunwoody, GA

- Migrated a backend Spring Boot service into a serverless Python application (using FastAPI, Pydantic) on AWS Lambda, reducing API latency by **40%**
- Containerized Python microservice with Docker & published production images to Amazon ECR, orchestrating workloads via Amazon ECS (Fargate) for horizontally scalable execution
- Configured AWS ECS to spin up Docker containers & exposed REST endpoints to React (S3) frontend via API Gateway, serving **100K+** requests monthly
- Cut deployment time by **95%+** by building a GitLab CI/CD pipeline & automating **200+** unit & integration tests

Georgia Tech - College of Computing

August 2024 – Present

Instructor for Introduction to Operating Systems

Atlanta, GA

- Built and deployed automated grading infrastructure (JUnit, pytest) backing a production backend system, executing **760K+ test runs** across **2,000+ students**
- Standardized and fixed CI pipelines (GitHub Actions, Gradle) across 25+ repos (150K+ LOC)
- Led weekly lecture with **100+** students, teaching: Networking Protocols (TCP/IP, DNS, HTTPS, BGP), Caching Protocols, Virtual Memory, Computer Architecture, Pipelining

Georgia Tech - Quantum Computing Lab

November 2023 – Present

Research Assistant

Atlanta, GA

- Created Quantum Playground, a browser-based interactive quantum computing environment for researchers to safely execute quantum code.
- Deployed backend on AWS EC2 with Ansible automation, Nginx load balancing, and Linux server configuration, handling **20K+ concurrent requests** with high availability via AWS CloudWatch.
- Built an interactive Rust-based REPL shell via PyO3, using memory-efficient Qubit encodings to educate researchers on traditional quantum algorithms

PROJECTS

Live Election Tracker | Python (Flask), Redis, Selenium, LeafletJS, React, AWS

- Built a full-stack election dashboard with Flask REST API + React, mapping 3K+ precincts via LeafletJS.
- Engineered a Selenium scraper aggregating 4M+ live votes daily with Redis caching for sub-second queries.
- Deployed scraper in prod, powering startup platform VoteHub and Kalshi betting markets with 2M+ daily visitors.

CheckPoint Grading | Python, Typescript, Docker, Java, C++

- Architected 15+ containerized autograding services (Docker, Python, Java) integrated with GitHub/CI pipelines, processing 200K+ submissions/year using isolated sandboxes & ML-based plagiarism detection.
- Designed a configurable grading engine with rubric versioning, test orchestration, automated feedback generation, and analytics dashboards; reduced manual grading time by 94%, saving hundreds of TA hours per semester.

TECHNICAL SKILLS

Languages: Java, Python, C/C++, Go, Rust, JavaScript, Kotlin, TypeScript, x86 Assembly, Swift

Frontend: React, Svelte, Next.js, Flutter, Figma, XCode, Jetpack Compose, CSS/HTML

Backend: AWS: Lambda, API Gateway, ECR, EC2, S3, Terraform; Node.js, Firebase, Nginx, FastAPI

Tools: Git/GitHub/GitLab, Docker, JUnit, Ansible, CI/CD, SSH, Kubernetes Bash/Shell Scripting

Other: Unix/Linux Environment, Bash, Multithreading, RESTful API, TCP/IP, VLANs, Agile Workflow, Compilers